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$$r = -\cos \theta$$



$$\begin{aligned} S &= 2 \cdot \frac{1}{2} \int_{\frac{\pi}{2}}^{\pi} \cos^2 \theta d\theta \\ \int \cos^2 \theta d\theta &= \int \frac{1 + \cos 2\theta}{2} d\theta = \frac{1}{2} \int d\theta + \frac{1}{2} \int \cos 2\theta d\theta = \frac{\theta}{2} + \frac{\sin 2\theta}{4} + c \\ \Rightarrow S &= \left[\frac{\theta}{2} + \frac{\sin(2\theta)}{4} \right]_{\frac{\pi}{2}}^{\pi} = \frac{\pi}{2} + \frac{\sin(2\pi)}{4} - \left(\frac{\pi}{4} + \frac{\sin(\pi)}{4} \right) = \frac{\pi}{2} - \frac{\pi}{4} = \frac{\pi}{4} \end{aligned}$$

References